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Architectural covering and method of manufacturing

Abstract

An architectural covering and a method of manufacturing the covering is provided. The panel may include multiple strips of material extending lengthwise across a width dimension of the panel. The strips of material may be overlapped and coupled to one another to define cells between adjacent strips of material. The panel may be retracted and extended across an architectural opening, and the strips of material may include a resilient support member to expand the cells as the panel is extended across the architectural opening. The panel may be manufactured by helically winding a continuous, elongate strip of material about a drum in an overlapped manner.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of co-pending U.S. patent application Ser. No. 16/843,136, filed Apr. 8, 2020, entitled “Architectural Covering and Method of Manufacturing”, which is a continuation of U.S. patent application Ser. No. 15/418,804, filed Jan. 30, 2017, now U.S. Pat. No. 10,648,229, entitled “Architectural

Covering and Method of Manufacturing”, which application claims the benefit of priority under 35 U.S.C. § 119(e) of the earlier filing date of U.S. Provisional Patent Application No. 62/357,237, filed 30 Jun. 2016 and entitled “Architectural Covering and Method of Manufacturing,” which are all hereby incorporated in their entireties.

FIELD

(1) The present disclosure relates generally to architectural coverings and methods of manufacturing architectural coverings, and more particularly to a panel of an architectural covering and a method of manufacturing the panel.

BACKGROUND

(2) Coverings for architectural structures or features (such as walls and openings, including windows, doors, archways, and the like) (hereinafter “architectural structure” for the sake of convenience without intent to limit) have assumed numerous forms for many years. Some coverings include a panel that defines multiple cells that trap air to increase the insulative factor of the covering. In some coverings, the panels are retractable or extendable across the architectural opening to alter the amount of light passage and visibility through the architectural opening. During retraction of the panel, the cells may collapse to decrease the volume of the cells, thereby providing a smaller panel to store along a side of the architectural opening. During extension of the panel, the cells may expand to increase the volume of the cells, thereby increasing the air trapped within the cells to increase the insulative factor of the panel.

(3) U.S. Patent Publication No. 2014/0053989 describes a panel including a support sheet and at least one cell operably connected to the support sheet. The at least one cell may include a vane material operably connected to a first side of the support sheet and a cell support member operably connected to the vane material and configured to support the vane material at a distance away from the support sheet when the panel is in an extended position with respect to the support tube.

(4) U.S. Patent Publication No. 2013/0105094 describes a process and system for manufacturing roller blinds which includes structure for performing plural steps including a first step of helically winding slat fabric about a drum, thereby forming a slat product. A second step includes moving the slat product from the drum to a platform. A third step includes winding the slat product about a roller tube to form a roller blind. A fourth step includes moving the blind from the platform to a heat treating device.

SUMMARY

(5) The present disclosure is at least partially directed to a panel and method of manufacturing a panel and generally provides a user with different panel and manufacturing options.

(6) The present disclosure generally provides an architectural covering including a panel and a method of manufacturing the panel. The panel may include multiple strips of material extending lengthwise across a width dimension of the panel. The strips of material may be overlapped and operably coupled to one another to define cells between adjacent strips of material. The panel may be retracted and extended across an architectural structure, and the strips of material may include a resilient support member to expand the cells as the panel is extended across the architectural structure. The panel may be manufactured by helically winding a continuous, elongate strip of material about a drum in an overlapped manner.

(7) The present disclosure is given to aid understanding, and one of skill in the art will understand that each of the various aspects and features of the disclosure may advantageously be used separately in some instances, or in combination with other aspects and features of the disclosure in other instances. Accordingly, while the disclosure is presented in terms of examples, it should be appreciated that individual aspects of any example can be claimed separately or in combination with aspects and features of that example or any other example.

(8) The present disclosure is set forth in various levels of detail in this application and no limitation as to the scope of the claimed subject matter is intended by either the inclusion or non-inclusion of

elements, components, or the like in this summary. In certain instances, details that are not necessary for an understanding of the disclosure or that render other details difficult to perceive may have been omitted. It should be understood that the claimed subject matter is not necessarily limited to the particular examples or arrangements illustrated herein.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The accompanying drawings, which are incorporated into and constitute a part of the specification, illustrate examples of the disclosure and, together with the general description given above and the detailed description given below, serve to explain principles of these examples.
- (2) FIG. 1 is an isometric view of an architectural covering in accordance with an embodiment of the present disclosure.
- (3) FIG. 2 is a fragmentary, enlarged isometric view of the covering of FIG. 1 in accordance with an embodiment of the present disclosure.
- (4) FIG. 3 is a transverse cross-sectional view of the covering of FIG. 1 taken along line 3-3 of FIG. 1 in accordance with an embodiment of the present disclosure.
- (5) FIG. 4 is an enlarged view of a first coupling line between adjacent strips of material and circumscribed in FIG. 3 in accordance with an embodiment of the present disclosure.
- (6) FIG. 5 is an enlarged view of a second coupling line between adjacent strips of material and circumscribed in FIG. 3 in accordance with an embodiment of the present disclosure.
- (7) FIG. 6 is an enlarged isometric view of a strip of material of the covering of FIG. 1 in accordance with an embodiment of the present disclosure.
- (8) FIG. 7 is an exploded view of the strip of material illustrated in FIG. 6 in accordance with an embodiment of the present disclosure.
- (9) FIG. 8 is a transverse cross-sectional view of the covering of FIG. 1 taken along line 8-8 of FIG. 1 in accordance with an embodiment of the present disclosure.
- (10) FIG. 9 is a flowchart of a method of manufacturing the covering of FIGS. 1-8 in accordance with an embodiment of the present disclosure.
- (11) FIG. 10 is an elevational view of a system for manufacturing the covering of FIGS. 1-8 illustrating a drum winding operation in accordance with an embodiment of the present disclosure.
- (12) FIG. 11 is a top plan view of the system of FIG. 10 in accordance with an embodiment of the present disclosure.
- (13) FIG. 12 is a longitudinal cross-sectional view of a panel of the covering of FIG. 1 taken along line 12-12 of FIG. 11 in accordance with an embodiment of the present disclosure.
- (14) FIG. 13 is an elevational view of the system of FIG. 10 illustrating a moving operation in accordance with an embodiment of the present disclosure.
- (15) FIG. 14 is a top plan view of the system of FIG. 10 illustrating a roller winding operation in accordance with an embodiment of the present disclosure.
- (16) FIG. 15 is an elevational view of the system of FIG. 10 illustrating a wound roller ready for heat treatment in accordance with an embodiment of the present disclosure.
- (17) FIG. 16 is an exploded view of the strip of material illustrated in FIG. 6 in accordance with an embodiment of the present disclosure.
- (18) FIG. 17 is a fragmentary, enlarged isometric view of the covering of FIG. 1 in accordance with an embodiment of the present disclosure.
- (19) FIG. 18 is a flowchart of a method of manufacturing the covering of FIG. 17 in accordance with an embodiment of the present disclosure.

DETAILED DESCRIPTION

- (20) In FIG. 1, the present disclosure illustrates an example of a covering 100 for an architectural

structure that includes a panel **102**. The panel **102** may include multiple strips of material **110** extending lengthwise across a width dimension of the panel **102**. The strips of material **110** may be overlapped and operably coupled to one another to define a cell between adjacent strips of material, thereby forming a cellular panel. The panel **102** may be retracted and extended across an architectural structure, and the strips of material **110** may include a resilient support member to expand the cells as the panel **102** is extended across the architectural structure. The panel **102** may be manufactured by helically winding a continuous, elongate strip of material about a drum in an overlapped manner.

(21) With continued reference to the illustrative embodiment illustrated in FIG. 1, the panel **102** may be retracted and extended across an architectural structure to adjust, for example, light transmission and/or visibility through the architectural structure. During retraction of the panel **102**, strips of material **110** of the panel **102** may generally collapse to decrease the volume of cells **106** formed by the overlapped strips of material **110**, thereby facilitating storage of the panel **102** along a side of the architectural structure. During extension of the panel **102**, strips of material **110** of the panel **102** may generally expand to increase the volume of cells **106** formed by the overlapped strips of material **110**, thereby increasing the air trapped within the cells **106** such as to increase the insulative factor of the panel **102**. The panel **102** may be configured so that at least a portion of each strip of material of the panel **102** may be biased to an expanded configuration as the panel **102** is extended across the architectural structure. In some embodiments, the strips of material **110** may be stacked upon one another and may extend laterally across the panel **102**. The cells **106** defined by the strips of material **110** may be enclosed along their length and may have open ends. The cells **106** may have various shapes, which may differ from that shown in the illustrated embodiments. Depending on the orientation of the covering **100**, the strips of material **110** may extend horizontally or vertically across the architectural opening.

(22) Referring to the illustrative embodiment of FIGS. 2, 3, and 8, the panel **102** may be manufactured without a support sheet, thereby reducing the cost and complexity of manufacturing the panel **102**. The panel **102** may include overlapping strips of material. Referring to the illustrative embodiment of FIGS. 2 and 3, a first elongated strip of material **110a** and an immediately adjacent second elongated strip of material **110b** may define an illustrative first cell **106a** therebetween, and the second elongated strip of material **110b** and an immediately adjacent third elongated strip of material **110c** may define an illustrative second cell **106b** therebetween. Referring specifically to first cell **106a** in FIG. 3 for illustrative purposes, the first strip of material **110a** and the second strip of material **110b** may overlap each other and may be coupled to each other along a first coupling line **114a** (see FIG. 4) and a second coupling line **114b** (see FIG. 5) to define cell **106a** therebetween. The first and second coupling lines **114a**, **114b** may extend lengthwise across the panel **102** and may be spaced apart from each other along a length of the panel **102** to define the first cell **106a** between the first and second strips of material **110a**, **110b**.

(23) Referring to FIGS. 2 and 3, each strip of material **110** may form a front wall of one cell and a rear wall of an immediately adjacent cell, thereby eliminating a support sheet for defining a rear wall of the cells. Referring specifically to the illustrative embodiment of FIG. 3, illustrative second strip of material **110b** may form a rear wall **118a** of first cell **106a** and a front wall **122b** of second cell **106b**. Illustrative first strip of material **110a** may form a front wall **122a** of first cell **106a**, and illustrative third strip of material **110c** may form a rear wall **118b** of second cell **106b**. A segment **124** of the second strip of material **110b** extending between rear wall **118a** and front wall **122b** may separate the first cell **106a** and the second cell **106b** from each other by defining a bottom wall **126a** of the first cell **106a** and a top wall **128b** of the second cell **106b**.

(24) Referring to FIG. 6, an illustrative embodiment of strip of material **110** is depicted. The illustrative strip of material **110** may include multiple creases or fold lines to facilitate collapse and expansion of the cells during retraction and extension, respectively, of the panel **102**. A first crease or fold line **134** (hereinafter “fold line” for the sake of convenience without intent to limit) may

extend lengthwise along a length of the strip of material **110**. The first fold line **134** may separate a curved or front wall portion **138** (hereinafter “curved portion” for the sake of convenience without intent to limit) from a substantially planar or rear wall portion **142** (hereinafter “substantially planar portion” for the sake of convenience without intent to limit) of the strip of material **110**. The curved portion **138** may form a front wall of a cell (for example front wall **122b** of second cell **106b** in FIG. 3), and the substantially planar portion **142** may form a rear wall of an adjacent cell (for example rear wall **118a** of first cell **106a** in FIG. 3). The first fold line **134** may function as a living hinge permitting the curved portion **138** to pivot about the first fold line **134** relative to the substantially planar portion **142**, thereby facilitating retraction and storage of the panel **102** (see FIG. 8).

(25) Referring to FIG. 3, the substantially planar portions **142** of adjacent strips of material **110** may be operably coupled together, thereby collectively forming a rear wall **146** of the panel **102** (see FIG. 8). As illustrated in FIG. 3, the substantially planar portion **142b** of second strip of material **110b** may overlap and be coupled to the substantially planar portion **142a** of the first strip of material **110a** along first coupling line **114a**, and the substantially planar portion **142c** of third strip of material **110c** may overlap and be coupled to the substantially planar portion **142b** of second strip of material **110b** along third coupling line **114c**, thereby forming a substantially planar rear wall **146** of the panel **102** (see FIG. 8). In the illustrative embodiment of FIG. 3, upper end portions of the substantially planar portions **142b**, **142c** of second and third strips of material **110b**, **110c** may overlap and be coupled to substantially planar portions **142a**, **142b** of first and second strips of material **110a**, **110b**, respectively, thereby forming stiffened regions adjacent first fold lines **134a**, **134b** of first and second strips of material **110a**, **110b**, respectively, which may facilitate pivoting of curved portions **138a**, **138b** about the first fold lines **134a**, **134b** relative to the substantially planar portions **142a**, **142b** of the first and second strips of material **110a**, **110b**, respectively, to collapse and expand cells **106a**, **106b**, respectively.

(26) Referring to FIG. 6, a second crease or fold line **154** (hereinafter “fold line” for the sake of convenience without intent to limit) may extend lengthwise along a length of the strip of material **110**. The second fold line **154** may separate the curved portion **138** from a tab **158** of the strip of material **110**. A third crease or fold line **156** (hereinafter “fold line” for the sake of convenience without intent to limit) may extend lengthwise along a length of the strip of material **110**. The third fold line **156** may be positioned between the second fold line **154** and the tab **158**, and in some embodiments may be positioned immediately adjacent the tab **158**. The tab **158** may form a portion of a bottom wall of a cell (for example bottom wall **126a** of first cell **106a** in FIG. 3). The second fold line **154** and the third fold line **156** may function as living hinges permitting the curved portion **138** to pivot about the second fold line **154** and the third fold line **156** relative to the tab **158**, thereby facilitating retraction and storage of the panel **102** (see FIG. 8).

(27) The tabs **158** of the strips of material **110** may be coupled to curved portions **138** of adjacent strips of material **110**, thereby collectively forming a front wall **162** of the panel **102** (see FIG. 8). As illustrated in FIG. 3, the curved portion **138a** of first strip of material **110a** may overlap the curved portion **138b** of second strip of material **110b**, and the tab **158a** of first strip of material **110a** may be coupled to the curved portion **138b** along second coupling line **114b**. Similarly, the curved portion **138b** of second strip of material **110b** may overlap the curved portion **138c** of third strip of material **110c**, and the tab **158b** of second strip of material **110b** may be coupled to the curved portion **138c** along fourth coupling line **114d**. The tabs **158a**, **158b** may be folded upwardly relative to the curved portions **138a**, **138b** along second folds **154a**, **154b** and third folds **156a**, **156b** and may be coupled to outer surfaces of the curved portions **138b**, **138c**, respectively. The overlapped curved portions **138** of the strips of material **110** may form a cascading front wall **162** of the panel **102** (see FIG. 8).

(28) Referring to FIG. 3, at least a portion of each curved portion **138** (for example curved portions **138a**, **138b**) may be visible from a front side of the panel **102** (see FIG. 8). At least these visible

portions of the curved portions **138** may include an aesthetic surface treatment (for example a color, texture, or other surface treatment) to enhance the aesthetics of the panel **102**. The curved portion **138** of the strip of material **110** may include a different surface treatment than the substantially planar portion **142**. For example, in some embodiments the curved portion **138** of the strip of material **110** may be colored and/or textured (such as by dyeing, printing, or other surface treatment methods), thereby providing an aesthetic front wall **162** of the panel **102** (see FIG. **8**) while reducing the cost of manufacturing the panel **102** by not applying the surface treatment to the entire strip of material **110** (e.g., to the substantially planar portion **142** of the strip of material **110**). In some embodiments, a layer of light-blocking material (hereinafter “blackout material” for the sake of convenience without intent to limit) may be applied to the strips of material **110** to inhibit light from being transmitted through the panel **102**. The blackout material may be applied to the curved portion **138**, the substantially planar portion **142**, or both. In some embodiments, the blackout material may be applied to a rear layer of the strips of material **110**. In one example, as illustrated in FIGS. **3-6**, a blackout material **164** may be applied to each strip of material **110**. The blackout material **164** of the strips of material **110** collectively may extend the full extent of the panel **102**, such that the blackout material **164** inhibits light from being transmitted through the panel **102** when the panel **102** is in a fully extended position. The blackout material **164** may be formed from various materials and thicknesses. In one example, the blackout material **164** may be formed of a non-woven film having light-blocking properties. The non-woven film may have a thickness that is less than about 5 mil. In one example, the film may have a thickness this is less than about 2 mil. In one example, the film may have a thickness that is greater than about one-half mil.

(29) Referring to FIG. **7**, each strip of material **110** may include multiple layers. For example, in the illustrative embodiment of FIG. **7**, the strip of material **110** may include a front layer **170**, a rear layer **174**, and an intermediate layer or cell support member **178** (hereinafter “cell support member” for the sake of convenience without intent to limit). The front layer **170** and the rear layer **174** may include substantially the same profile as each other and as the strip of material **110**. The front layer **170** and the rear layer **174** may be formed from a fabric material, such as a nonwoven fabric material. The front layer **170** and the rear layer **174** may be formed from the same material or different materials. In some embodiments, the front layer **170** and the rear layer **174** are formed from different nonwoven fabric materials. For example, the rear layer **174** may be formed from a less expensive material than the front layer **170** in examples where the rear layer **174** is not visible from a room side of the covering **100**.

(30) Referring to FIGS. **4-7**, the blackout material **164** may extend along the front layer **170** and/or the rear layer **174** for a majority of the distance between the top of the front and rear layers **170**, **174** and the first fold **134**. As illustrated in FIG. **6** (with reference to FIGS. **3-5**), the blackout material **164** may extend from aligned upper edges of the front and rear layers **170**, **174** and may terminate adjacent the first fold **134**. In the example illustrated in FIGS. **4** and **6**, the blackout material **164** may extend beyond the first fold **134** and may terminate adjacent the cell support member **178**. As illustrated in FIG. **3**, the blackout material **164** of adjacent strips of material **110** may overlap one another along a vertical extent of the rear wall **146** of the panel **102** to inhibit light from being transmitted through the panel **102**. In one example, the blackout material **164** extends past the first fold **134** by about one-eighth of an inch. As illustrated in FIGS. **6** and **7**, the blackout material **164** may be positioned (e.g., sandwiched) between the front and rear layers **170**, **174**. In the example illustrated in FIGS. **6** and **7**, the blackout material **164** may be spaced a distance from the cell support member **178** to provide a gap to account for manufacturing tolerance/variance of the dimensions of the blackout material **164** and the cell support member **178** to ensure the blackout material **164** and the cell support member **178** do not overlap, which overlap may cause an undesirable ripple or other surface irregularity visible from a front side of the panel **102**. The blackout material **164** may be coupled with the front and rear layers **170**, **174** by adhesive, heat

sealant, or other techniques.

(31) Referring to FIGS. 6 and 7, the cell support member **178** may be positioned (e.g., sandwiched) between the curved portions **138** of the front and rear layers **170**, **174** and may extend along the front and rear layers **170**, **174** for a majority of the distance between the first fold **134** and the second fold **154**. As illustrated in FIG. 6 (with reference to FIG. 5), the cell support member **178** may extend from the second fold **154** to or beyond the coupling line **114b** (see FIG. 5) and may terminate adjacent the first fold **134**. In the illustrative embodiment of FIG. 5, the cell support member **178** of the strip of material **110b** extends beyond the coupling line **114b**, such that the cell support member **178** facilitates transfer of a biasing force from the superjacent strip of material **110a** to the strip of material **110b** to provide additional cantilever for the strip of material **110b**. The cell support member **178** may terminate between the coupling line **114** and the first fold **134** (see, for example, coupling line **114b** and first fold **134b** in FIGS. 3 and 5). In some embodiments, the cell support member **178** may terminate a distance from the first fold **134**. For example, the cell support member **178** may terminate less than about two inches from the first fold line **134**. In some examples, the cell support member **178** may terminate less than about one inch from the first fold line **134**. In some examples, the cell support member **178** may terminate about 0.375 inches from the first fold line **134**, though other distances may be used depending on the particular application. The cell support member **178** may include substantially the same curvature as the curved portion **138** of the front and rear layers **170**, **174**. The cell support member **178** may stiffen and/or optionally impart curvature to the curved portion **138** of the strip of material **110** and may be resilient to facilitate expansion of the respective cell of the panel **102** from a collapsed configuration. As illustrated in FIG. 7, third fold lines **156** may be formed between the second fold lines **154** and the tabs **158** in both the front and rear layers **170**, **174**. The third fold lines **156** may be spaced a distance apart from the second fold lines **154** and may define an inner edge of the tabs **158**. The tabs **158** may be used to couple a respective strip of material **110** to a subjacent strip of material **110**.

(32) Referring to FIG. 3, each strip of material **110** may include a support member **178** that is resilient so as to allow the strips of material **110**, and thus the cells **106**, to at least partially collapse when the panel **102** is retracted, and spring or bias to an expanded configuration when the panel **102** is extended. A “collapsed” cell includes the configuration where the front and rear walls of a respective cell are positioned adjacent each other (e.g., in contact or in partial contact), and an “expanded” cell includes the configuration where the front and rear walls of a respective cell are spaced from each other to define an insulative air chamber or void between the front and rear walls.

(33) The cell support member **178** may be a thermoformable material that becomes partially or substantially shapeable after heating, and retains its formed shape after cooling. The cell support member **178** may be a moldable film, such as polyester film, or other thermoformable material. The cell support member **178** may have an adhesive-like property when heated and then cooled. The cell support member **178** may be coupled to the front and rear layers **170**, **174** by adhesion, stitching, ultrasonic welding, or other coupling techniques or methods. In some embodiments, the cell support member **178** may be adhered to the front and rear layers **170**, **174** with an adhesive that sets at a temperature below the forming temperature of the cell support member **178**, thereby permitting coupling of the cell support member **178** to the front and rear layers **170**, **174** in a substantially planar configuration and subsequent thermoforming of the cell support member **178** to set a spiral curvature of the cell support member **178** and thus the curved portion **138** of the strip of material **110** (see FIGS. 6 and 7).

(34) To set the spiral curvature of the cell support member **178**, the strip of material **110** may be wound around a support tube, mandrel, or other forming member and then heated. As the components are heated, the cell support member **178** may be re-shaped to conform generally to the shape of the forming member. After cooling, the curved portion **138** of the strip of material **110** may have the shape of the cell support member **178**. A method of manufacturing the panel **102** is

described in more detail below.

(35) Referring to FIG. 8, the covering **100** may include a head rail **184** and an end rail **186**. A roller **188**, such as a support tube, may be positioned in the head rail **184**, and the panel **102** may be coupled to the roller **188** for retracting and extending the panel **102** across the architectural opening as the roller is rotated in a selected direction. The end rail **186** may be coupled to the panel **102** opposite the roller **188**, and the weight of the end rail **186** may tension the panel **102** when extended to help expand the strips of material **110**, and thus the cells **106**, from a collapsed configuration to an expanded configuration.

(36) As shown in the illustrative embodiment of FIG. 8, the panel **102** may be wound around the roller **188**. As the panel **102** is wound around the roller **188**, the effective length of the panel **102** decreases and the end rail **186** is moved towards the head rail **184**. The head rail **184** may be dimensioned to house or receive substantially the entire panel **102** wound around the roller **188**, such that the panel **102** may be substantially hidden from view within the head rail **184**, which may provide protection from ultra-violet sunlight damage, dust, and other environmental factors. The end rail **186** may be received through an opening in an underside of the head rail **184**, or may abut against the underside of the head rail **184** when the panel **102** is in a fully retracted position.

(37) During retraction of the panel **102**, the strips of material **110** may collapse to decrease the volume of the cells **106**, thereby decreasing a depth distance between overlapped portions of adjacent strips of material to facilitate storage of the panel **102** along a side of the architectural opening, such as wrapped around the roller **188** within the head rail **184**. The strips of material **110** may collapse when wound around the roller **188** because, for example, the first and second folds **134**, **154** of the strips of material **110** (see FIG. 7) may allow the strips of material **110** to deform into a wound configuration having a curvature that generally corresponds to the curvature of the cell support members **178**, which may be substantially equal to a curvature of the roller **188**. During collapse of the strips of material **110**, the substantially planar portions **142** of the strips of material **110** may conform to the curved portion **138** of adjacent strips of material **110**, and as previously discussed the curved portion **138** may have a curvature that generally corresponds to the curvature of the cell support member **178**.

(38) During extension of the panel **102**, the strips of material **110** may expand to increase the volume of the cells **106**, thereby increasing the air trapped between adjacent strips of material **110** to increase the insulative factor of the panel **102**. The panel **102** may be configured so that each strip of material **110** within the panel **102** may be biased to expand as the panel **102** is extended to ensure each cell **106** fully expands during extension to increase the insulative factor of the panel **102** and provide a uniform appearance along the length of the panel **102**. For example, as the panel **102** is unwound from the roller **188**, the cell support member **178** of each strip of material **110** may bias the cells **106** toward an expanded configuration. The cell support members **178** and the first and second folds **134**, **154** of the strips of material **110** (see FIG. 7) may apply tension to the substantially planar portions **142** of the strips of material **110** to remove slack in the substantially planar portions **142**, thereby moving the substantially planar portions **142** away from corresponding curved portions **138** of respective cells **106** to expand the strips of material **110** and increase the insulative factor of the panel **102**.

(39) Referring to FIG. 9, a method **200** of manufacturing an architectural covering is illustrated. The method **200** may include helically winding an elongated strip of material about a drum to form a panel (operation **204**). The method **200** may include moving the panel from the drum to a platform (operation **208**). The method **200** may include winding the panel about a roller to form a wound roller (operation **212**). The method **200** may include heat treating the wound roller to set a spiral curvature into the elongated strip of material (operation **216**). The method **200** may be synchronized, so that a first-formed covering product may be moved from the platform to a heat treating device, substantially when a second-formed covering product is moved from the drum to the platform.

(40) Turning now to FIGS. 10 and 11, a system 300 capable of performing operation 204 of FIG. 9 is illustrated. As illustrated in FIGS. 10 and 11, system 300 may include a drum 302 that is rotatable about its center axis such as by suitable automated machinery. The drum 302 may have axial front and rear ends 304, 306 (see FIG. 11). The drum 302 may include an opening or groove 308 (hereinafter “groove” for the sake of convenience without intent to limit), which may extend the axial length of the drum 302 defined between the front and rear ends 304, 306. The drum 302, which may be formed as a shell, may have a circumference defining a width dimension of a wound panel, which subsequently may be subdivided into smaller width panels. In some embodiments, the circumference of the drum 302 may be about sixteen feet. The axial length of the drum 302 may define a length of the wound panel, which subsequently may be subdivided into smaller length panels.

(41) Referring still to FIGS. 10 and 11, the system 300 may include a distributing structure 310 for distributing material against an outer surface of the drum 302. The distributing structure 310 may include a supply roll 312 of an elongated strip of material 314 (such as strip of material 110 in FIG. 6 in a pre-folded/molded configuration as illustrated in exploded form in FIG. 16) for winding around the drum 302. The distributing structure 310 may include an adhesive dispenser 316 for applying adhesive 318 to the elongated strip of material 314 prior to the elongated strip of material 314 being wound around the drum 302. The adhesive dispenser 316 may apply multiple lines of adhesive 318 (such as coupling lines 114a and 114b in FIG. 3) to the elongated strip of material 314, and the lines of adhesive 318 may be spaced apart from each other along a width of the strip of material 314. The lines of adhesive 318 may be applied adjacent first and second folds of the strip of material 314 (such as first and second folds 134, 154 of the strip of material 110 in FIG. 3). The lines of adhesive 318 may adhere adjacent layers or windings (hereinafter “layers” for the sake of convenience without intent to limit) of the strip of material 314 to one another to define cells (such as cells 106 in FIG. 3) therebetween.

(42) The distributing structure 310 may include a folding structure 320 for folding the strip of material 314 (for example, for folding the tab 158 along fold line 154 of the strip of material 110 in FIG. 3) such that adhesive may be applied to the tab 158. The folding structure 320 may be formed as an open-ended box with an inlet having a larger width dimension than an outlet of the box such that the strip of material 314 enters the inlet of the box and at least one of the side walls of the box tapers inwardly from the inlet to the outlet of the box to fold the strip of material 314 along its width (such as folding the tab 158 along fold line 154 of the strip of material 110 in FIG. 3). The distributing structure 310 may include one or more guide rollers 322, 324 for guiding the strip of material 314 from the supply roll 312 through the folding structure 320 to the adhesive dispenser 316 and onto the drum 302.

(43) The strip of material 314 may be helically wound around the drum 302 such that each winding of the strip of material 314 about the drum 302 may form a strip of material 110 of the panel 102 of FIG. 1. As illustrated in FIG. 12, the panel 342 may be relatively flat when wound around the drum 302 and may form the panel 102 of FIG. 1 after subsequent operations discussed below. A front wall 162 of panel 102 may face inwardly toward the drum 302, and a rear wall 146 of panel 102 may face outwardly away from the drum 302. The strips of material 110 may be oriented such that front walls 122 of cells 106 may be wound against the drum 302 and rear walls 118 may face outwardly away from the drum 302. The tabs 158 of strips of material 110 may be folded over by the folding structure 320 (see FIG. 10) such that the tabs 158 may face outwardly away from the drum 302. The lines of adhesive 318 applied by the adhesive dispenser 316 (see FIG. 10) may form coupling lines 114 for coupling overlapped strips of material 110, and the lines of adhesive 318 may face outwardly away from the drum 302 for coupling to an adjacent, overlapped winding of the strip of material 314. In some embodiments, front wall 162 of the panel 102 may face outwardly away from the drum 302, and rear wall 146 of the panel 102 may face inwardly toward the drum 302.

(44) Referring to FIG. 11, the distributing structure 310 may move axially along a side of the drum 302 during winding of the strip of material 314 around the drum 302. The distributing structure 310 may be initially positioned near the axial front end 304 of the drum 302 and may move in an axial direction 328 toward the rear end 306 of the drum 302 during the winding process. As illustrated in FIG. 10, when winding the strip of material 314 about the drum 302, the drum 302 may spin in a clockwise direction 330. The distributing structure 310 may automatically travel towards the axial rear end 306 of the drum 302 during rotation of the drum 302, resulting in the strip of material 314 being helically wound around the drum 302. The rate of axial advancement of the distributing structure 310 may be based on the desired overlap of adjacent strips of material 110 on the panel 102 (see FIGS. 2 and 3). A faster rate of axial advancement may reduce the overlap resulting in a smaller cell size (e.g., reduced cell height and/or depth) and a slower rate of axial advancement may increase the overlap resulting in a larger cell size (e.g., increased cell height and/or depth). Adjustment of the rate of axial advancement of the distributing structure 310 may be automated by computer-based controls. The distributing structure 310 may be capable of translating in either axial direction between the front end 304 and the rear end 306 of the drum 302.

(45) Referring to FIG. 10, the winding process may wrap the strip of material 314 continuously around the drum 302 such that the strip of material 314 extends across the axial groove 308 in the drum 302. As illustrated in FIGS. 10 and 11, a cutter 334, such as a rotary cutting wheel, may be supported on a gantry system 336, for example. Once a desired amount of strip of material 314 is wound about the drum 102, the drum 102 may be rotated to align the axial groove 308 with the cutter 334, such as positioning the axial groove 308 along a top of the drum 302. Once aligned, the cutter 334 may extend within the groove 308 and travel axially along the drum 102 between the opposing front and rear ends 304, 306 of the drum 302 (see arrow 340 in FIG. 11) to cut the wound strip of material 314 and form a panel 342 with side edges (see, e.g., panel 102 in FIG. 1). One of the edges of the panel 342 may be held against the drum 102 along one side of the axial groove 308 by one or more magnets, for example.

(46) As illustrated in FIGS. 11 and 13, the platform 346 may take the form of a table or workbench, which may have a rectangular working surface sufficiently large to support the panel 342. A right side edge 350 of the platform 346 may be disposed above a portion of a left side of the drum 302 adjacent to the axial groove 308 in the drum 302. The positional relationship between the platform 346 and the drum 302 may facilitate an efficient transfer of the panel 342 from the drum 302 onto the platform 346. If a front wall of the panel 342 is wound against the drum 302 during the drum winding process, then the panel 342 may be transferred directly to the platform 346 without flipping over the panel 342, thereby facilitating an efficient transfer of the panel 342 from the drum 302 onto the platform 346.

(47) Referring to FIG. 13, the system 300 may move the panel 342 from the drum 102 to a platform 346. For example, the system 300 may grip a side edge of the panel 342 formed by the cutter 334 and may move the gripped edge lateral to the axis of the drum 302 to unwind the panel 342 from the drum 302. As illustrated in FIG. 13, a leading edge 352 of the panel 342 may be sandwiched by a metallic bar 354, and a magnet 356 (including magnetic or ferrous materials) coupled to the gantry system 336 may magnetically interact with the metallic bar 354 to thereby grip the leading edge 352 of the panel 342. The gantry system 336 may be configured to move the magnet 356, and thus the metallic bar 354 and the leading edge 352 of the panel 342, lateral to the axis of the drum 302 across the platform 346 (see arrow 358 in FIG. 12) to unwrap the panel 342 from the drum 302. Once the panel 342 is fully supported by the platform 346, the gantry system 336 may be configured to release the magnetic interaction between the magnet 356 and the metallic bar 354, thereby releasing the leading edge 352 of the panel 342. Depending on its desired dimensions, the panel 342 may be trimmed along its edges once positioned on the platform 346. The trimming operation may ensure the panel 342 is rectangular. The trimming operation may be performed by a manual or automated cutter (not illustrated).

(48) Referring to FIG. 13, the system 300 may wind the panel 342 about a roller (such as roller 188 in FIG. 8) to form a wound roller. As illustrated in FIG. 13, the system 300 may include a supply of rollers 360. The system 300 may obtain a roller 362 from the supply 360 and position the roller 362 against a top edge portion 364 of the rear surface 366 of the panel 342. To create the proper alignment between the roller 362 and the panel 342, the roller 362 may be positioned at an angle equivalent to an angular offset at which the elongated strip of material 314 is helically wound about the drum 302. Alternatively, the panel 342 may be re-oriented on the platform 346 by the gantry system 336, for example, such that the panel 342 is square to the platform 346, and thus the roller 362 may be positioned substantially parallel to a front edge of the platform 346 and the top edge portion 364 of the panel 342. The trimming operation previously discussed may ensure the strips of material of the panel 342 extend substantially perpendicular to the side edges of the panel 342. The roller 362 may be coupled to the top edge portion 364 of the panel 342 by pre-coating the roller 362 with double sided tape (not illustrated) or applying a layer of double sided tape to the top edge portion 364 of the panel 342, for example.

(49) As illustrated in FIG. 14, a roller motor 370 may engage an end 372 of the roller 362. The roller motor 370 may turn the roller 362, thereby wrapping the panel 342 about the roller 362 to provide a curvature on each strip of material 110 (see FIG. 6), which may be defined by a segment of the spiral curve on which the cell support member 178 (see FIG. 6) is positioned when the panel 342 is wound about the roller 362. When wound around the roller 362, the panel 342 may be encapsulated, such as by a sheet of material 374 (see FIG. 15), to keep the panel 342 tightly wound about the roller 362 during subsequent processing, such as cutting in a rotary cutting process, storage, or other processing. The sheet of material 374 (hereinafter “tail paper” for the sake of convenience without intent to limit) may be dimensioned to wrap around the full circumference of the panel 342 that is wound about the roller 362 to protect the panel 342 from damage, such as from dirt or other debris. In some embodiments, the tail paper 374 may be dimensioned such that it may wrap around the full circumference of the panel 342 at least one time, such as about 1.25 to 1.33 times, or other numbers of times depending on the particular application. An upper edge portion of the tail paper 374 may be coupled to a bottom edge portion of the panel 342 (coupling not shown in FIG. 15) in various manners, such as via a strip of pressure sensitive tape. A lower edge portion of the tail paper 374 may be coupled to a previous winding of the tail paper 374 and may include an alternating assembly of tear strips and pressure sensitive tape to permit coupling of the tail paper 374, later inspection of the panel 342, and subsequent re-coupling of the tail paper 374. For example, the lower edge portion of the tail paper 374 may include a first tear strip, a first pressure sensitive tape, a second tear strip, and a second pressure sensitive tape spaced along the lower edge portion of the tail paper 374. The alternating arrangement may allow an operator to couple the lower edge portion of the tail paper 374 to a previous winding of the tail paper 374 along a lowermost-arranged pressure sensitive tape, and subsequently open the tail paper 374 for inspecting the panel 342 by pulling on an adjacent tear strip. After inspection, the operator may remove a release film from an adjacent pressure sensitive tape and re-roll the panel 342 about the roller 362, thereby encapsulating the tail paper 374 around the panel 342 and securing it in place via the adjacent pressure sensitive tape. The wound roller may be packaged tightly for storage, cutting, or other processing. The tail paper 374 may be formed from various materials and may include branding information (which may be printed on the tail paper) to facilitate identification of the type of panel, for example. The tail paper 374 may include a scale (which may be printed on the tail paper) extending along a length dimension of the roller 362. The scale may facilitate cutting the panel 342 to a desired width. The scale may facilitate quick identification of the width of a wound panel 342, such as when an operator is looking for a wound panel including a desired width amongst a stock of stored wound panels.

(50) Referring to FIG. 15, the system 300 may apply heat to the wound roller 376 to set a curvature into respective cells of the panel 342 (such as by thermoforming the cell support member 178 to set

the curvature of the curved portion **138** of the strip of material **110** in FIG. 6). A heat treating device, such as an oven **378**, may be positioned under the platform **346** and an inlet to the oven **378** may be positioned near the area for winding the panel **342** around the roller **362** such that the wound roller **376** may be automatically fed into the inlet of the oven **378**. The system **300** may include a pivot arm **380** configured to pivot a section **382** of the platform **346** in a downward direction, as illustrated in FIG. 15, and allow gravity to feed the wound roller **376** into an inlet of the oven **378**. In the oven **378**, the wound roller **376** may be heat treated so that the spiral curvature in each cell support member **178** (see FIG. 6) is permanently set. For example, the cell support member **178** may be formed from material that is thermoformable above about 170 degrees F. and/or below about 250 degrees F. In this example, the oven **378** may heat the wound roller **376** above about 170 degrees F. and/or below about 250 degrees F. to set a spiral curvature profile into the cell support members **178** (see FIG. 6). The oven **378** may be a standard convective type or a different type of oven which is capable of activating the thermoformable properties within the cell support members **178** (see FIG. 6). As the heat treatment process may be substantially longer than the time to form a wound roller **376**, the oven **378** may be sufficiently large to hold multiple wound rollers so that the system **300** may continuously heat treat the wound rollers. A system similar to system **300** is described in U.S. Patent Publication Number 2013/0105094 to Colson et al., entitled "Process and System for Manufacturing a Roller Blind", which publication is incorporated herein by reference in its entirety.

(51) Referring to the illustrative embodiment of FIG. 17, the panel may optionally be manufactured with a separate support sheet formed from multiple strips of material. In the following description, elements or components similar to those in the embodiment of FIGS. 1-8 are designated with the same reference numbers increased by 100 and redundant description is omitted. As illustrated in FIG. 17, the panel **202** may include a first set of overlapping strips of material **210** forming a front sheet or wall of the panel **202** and a second set of overlapping strips of material **211** forming a rear sheet or wall of the panel **202**. In the illustrative embodiment of FIG. 17, first, second, and third elongated overlapping strips of material **210a**, **210b**, **210c** may be coupled together along tabs **258** via, for example, adhesive, stitching, or other techniques to form a portion of the front wall of the panel **202**. First, second, and third elongated overlapping strips of material **211a**, **211b**, **211c** may be coupled together along their overlaps via, for example, adhesive, stitching, or other techniques to form a corresponding portion of the rear wall of the panel **202**. The first and second sets of overlapping strips of material **210**, **211** may be coupled together along the strips of material **210** above the fold lines **234** via, for example, adhesive, stitching, or other techniques, to form cells **206** between the first and second sets of overlapping strips of material **210**, **211**. In the illustrative embodiment of FIG. 17, first strips of material **210a**, **211a** may define an illustrative first cell **206a** therebetween, and second strips of material **210b**, **211b** may define an illustrative second cell **206b** therebetween. A panel similar to panel **202** is described in U.S. Patent Publication Number 2014/0053989 to Colson et al., entitled "Covering for Architectural Opening including Cell Structures Biased to Open", which publication is incorporated herein by reference in its entirety.

(52) Referring to FIG. 18, a method **300** of manufacturing an architectural covering is illustrated. In the following description, elements or components similar to those in the embodiment of FIGS. 9-15 are designated with the same reference numbers increased by 100 and redundant description is omitted. The method **300** may include helically winding multiple elongated strips of material about a drum to form a panel (operation **304**). The method **300** may include moving the panel from the drum to a platform (operation **308**). The method **300** may include winding the panel about a roller to form a wound roller (operation **312**). The method **300** may include heat treating the wound roller to set a spiral curvature into the elongated strip of material (operation **316**). The method **300** may be synchronized, so that a first-formed covering product may be moved from the platform to a heat treating device, substantially when a second-formed covering product is moved from the drum to the platform.

(53) Operation **304** of FIG. **18** may be performed by system **300** illustrated in FIGS. **10** and **11**, which operates as previously described in relation to operation **204** of FIG. **9**, except the distributing structure **310** distributes first and second elongated strips of material (such as strips of material **210**, **211** in FIG. **17**) in subsequent passes along a length of the drum **302**. The supply roll **312** of the distributing structure **310** may include an elongated strip of material **314** (initially, e.g., strip of material **210** in FIG. **17**) for winding first around the drum **302**. The adhesive dispenser **316** of the distributing structure **310** may apply adhesive **318** to the strip of material **210** (see FIG. **17**) prior to the strip of material **210** being wound around the drum **302**. The adhesive dispenser **316** may apply multiple lines of adhesive **318** (such as coupling lines **314a** and **314b** in FIG. **17**) to the strip of material **210** (see FIG. **17**), and the lines of adhesive **318** may be spaced apart from each other along a width of the strip of material **210**. The lines of adhesive **318** may be applied adjacent first and second folds of the strip of material **210** (such as first and second folds **234**, **254** in FIG. **17**). One of the lines of adhesive **318** (such as coupling line **314b** in FIG. **17**) may adhere adjacent layers or windings (hereinafter “layers” for the sake of convenience without intent to limit) of the strip of material **210** (see FIG. **17**) to one another to define a panel wall (such as front wall **210** of panel **202** in FIG. **17**).

(54) Referring to FIG. **11**, the distributing structure **310** may move axially along a side of the drum **302** during winding of the strip of material **314** (e.g., strip of material **210** in FIG. **17**) around the drum **302**, resulting in the strip of material **210** being helically wound around the drum **302**. The rate of axial advancement of the distributing structure **310** may be based on the desired overlap of adjacent strips of material **210** on the panel **202** (see FIG. **17**). The distributing structure **310** may be capable of translating in either axial direction between the front end **304** and the rear end **306** of the drum **302**.

(55) Referring to FIGS. **10** and **11**, after the strip of material **314** (e.g., strip of material **210** in FIG. **17**) is applied to the outer surface of the drum **302** by the distributing structure **310**, the supply roll **312** of the distributing structure **310** may be furnished with another elongated strip of material **314** (e.g., strip of material **211** in FIG. **17**, which may be the same or a different material than strip of material **210** in FIG. **17**) for winding around the drum **302** on top of the strip of material **210** already wound onto the drum **302**. The distributing structure **310** may move axially along a side of the drum **302** during winding of the strip of material **211** (see FIG. **17**) around the drum **302** onto the strip of material **210**, resulting in the strip of material **211** being helically wound around the drum **302** onto the strip of material **210**. One of the lines of adhesive **318** (such as coupling lines **314a** in FIG. **17**) may adhere the strips of material **210**, **211** (see FIG. **17**) together during the first and second passes of the strips of material **314**. The adhesive dispenser **316** of the distributing structure **310** may apply another line of adhesive **318** (e.g., line of adhesive **314c** in FIG. **17**) to the strip of material **211** (see FIG. **17**) to adhere adjacent layers of the strip of material **211** (see FIG. **17**) to one another to define a rear wall **211** of panel **202** in FIG. **17**. Thus, the adhesive dispenser **316** may apply three lines of adhesive (e.g., lines of adhesive **314a**, **314b**, **314c** in FIG. **17**) during the first and second passes of the strips of material **314** (e.g., strips of material **210**, **211** in FIG. **17**). The distributing structure **310** may translate in the same axial direction or different axial directions between the front end **304** and the rear end **306** of the drum **302** to apply the first and second passes of the elongated strips of material **314** (e.g., strips of material **210**, **211** in FIG. **11**).

(56) Referring to FIG. **18**, to continue manufacturing the panel **202** of FIG. **17**, operations **308**, **312**, **316** of method **300** may be performed. Operations **308**, **312**, **316** of method **300** are substantially the same as operations **208**, **212**, **216** of method **200** previously described in relation to FIGS. **9-15**. Thus, operations **308**, **312**, **316** will not be further described here.

(57) The discussion of any embodiment is meant only to be explanatory and is not intended to suggest that the scope of the disclosure, including the claims, is limited to these examples. In other words, while illustrative embodiments of the disclosure have been described in detail herein, it is to be understood that the inventive concepts may be otherwise variously embodied and employed, and

that the appended claims are intended to be construed to include such variations, except as limited by the prior art.

(58) The foregoing discussion has been presented for purposes of illustration and description and is not intended to limit the disclosure to the form or forms disclosed herein. For example, various features of the disclosure are grouped together in one or more aspects, embodiments, or configurations for the purpose of streamlining the disclosure. However, it should be understood that various features of the certain aspects, embodiments, or configurations of the disclosure may be combined in alternate aspects, embodiments, or configurations. Moreover, the following claims are hereby incorporated into this Detailed Description by this reference, with each claim standing on its own as a separate embodiment of the present disclosure.

(59) The phrases “at least one”, “one or more”, and “and/or”, as used herein, are open-ended expressions that are both conjunctive and disjunctive in operation. The term “a” or “an” entity, as used herein, refers to one or more of that entity. As such, the terms “a” (or “an”), “one or more” and “at least one” can be used interchangeably herein.

(60) All directional references (e.g., proximal, distal, upper, lower, upward, downward, left, right, lateral, longitudinal, front, back, top, bottom, above, below, vertical, horizontal, radial, axial, clockwise, and counterclockwise) are only used for identification purposes to aid the reader's understanding of the present disclosure, and do not create limitations, particularly as to the position, orientation, or use of this disclosure. Connection references (e.g., attached, coupled, connected, and joined) are to be construed broadly and may include intermediate members between a collection of elements and relative movement between elements unless otherwise indicated. As such, connection references do not necessarily infer that two elements are directly connected and in fixed relation to each other. Identification references (e.g., primary, secondary, first, second, third, fourth, etc.) are not intended to connote importance or priority, but are used to distinguish one feature from another. The drawings are for purposes of illustration only and the dimensions, positions, order and relative sizes reflected in the drawings attached hereto may vary.

Claims

1. An architectural covering, the covering comprising: a rotatable roller; a panel coupled to the roller so that rotation of the roller moves the panel between a retracted position and an extended position, the panel being wound around the roller in the retracted position, the panel comprising: a first elongated strip of material including a first portion and a second portion separated from the first portion by a first fold; and a second elongated strip of material including a third portion and a fourth portion separated from the third portion by a second fold, the second elongated strip of material overlapped with and coupled to the first elongated strip of material; wherein the first and second elongated strips of material define an enclosed cell; wherein each of the first elongated strip of material and the second elongated strip of materials is manufactured from a plurality of layers including a front layer, a rear layer, and an intermediate layer disposed between the front layer and the rear layer; wherein for each of the first portion of the first elongated strip of material and the third portion of the second elongated strip of material, the intermediate layer comprises a support member such that the first portion of the first elongated strip of material and the third portion of the second elongated strip of material include the front layer, the rear layer, and the support member; and wherein the second portion of the first elongated strip of material and the fourth portion of the second elongated strip of material includes the front layer and the rear layer and does not include the support member.

2. The architectural covering of claim 1, wherein the intermediate layer further includes a blackout material, the second portion of the first elongated strip of material and the fourth portion of the second elongated strip of material includes the blackout material, the first portion of the first elongated strip of material and the third portion of the second elongated strip including the support

member and does not include the blackout material

3. The architectural covering of claim 1, wherein the support member comprises a thermoformable material.

4. The architectural covering of claim 1, wherein: the first portion forms a front wall of the enclosed cell; a portion of the third portion forms a bottom wall of the enclosed cell; and the fourth portion forms a rear wall of the enclosed cell.

5. The architectural covering of claim 1, wherein: the first portion and the third portion are visible from a front side of the panel in the extended position; and the second portion and the fourth portion are hidden from the front side of the panel in the extended position.

6. The architectural covering of claim 1, wherein: the fourth portion overlaps and is coupled to the second portion; and the first portion overlaps a portion of the third portion.

7. The architectural covering of claim 6, wherein: the first portion transitions into a first tab that is coupled to the portion of the third portion of the second elongated strip of material; and the first portion is separated from the first tab by a third fold.

8. The architectural covering of claim 7, wherein: the fourth portion is coupled to the second portion along a first coupling line extending an entire length of the first elongated strip of material; and the first tab is coupled to the portion of the third portion along a second coupling line extending an entire length of the first elongated strip of material.

9. The architectural covering of claim 8, wherein: the first coupling line comprises a first line of adhesive; and the second coupling line comprises a second line of adhesive.

10. The architectural covering of claim 1, wherein: the first portion and the third portion include a curved profile; and the second portion and the fourth portion are planar.

11. An architectural covering, the covering comprising: a rotatable roller; a panel coupled to the roller so that rotation of the roller moves the panel between a retracted position and an extended position, the panel being wound around the roller in the retracted position, the panel comprising: a front sheet including a first set of overlapping strips of material; and a rear support sheet including a second set of overlapping strips of material coupled to each other; wherein: the second set of overlapping strips of material include at least first and second elongated overlapping strips of material, each of the first and second elongated overlapping strips of material including a top end portion and a bottom end portion, the bottom end portion of the first elongated overlapping strip of material overlapping with the top end portion of the second elongated overlapping strip of material; the first set of overlapping strips of material include at least first and second elongated overlapping strips of material, each of the first and second elongated overlapping strips of material of the first set of overlapping strips of material including a first portion and a second portion separated from the first portion by a first fold; the first portion of the first elongated overlapping strip of material of the first set of overlapping strips of material is coupled to a bottom end portion of the first elongated overlapping strip of material of the second set of overlapping strips of material; the second portion of the first elongated overlapping strip of material of the first set of overlapping strips of material is coupled to a first portion of the second elongated overlapping strip of material of the first set of overlapping strips of material; wherein each of the first and second elongated overlapping strips of material of the first set of overlapping strips of material is manufactured from a plurality of layers including a front layer, a rear layer, and an intermediate layer disposed between the front layer and the rear layer; and wherein the intermediate layer of the first portion of the first overlapping strip of material of the first set of overlapping strips of material comprises a support member so that the first portion includes the front layer, the rear layer, and the support member, the second portion of the first overlapping strip of material of the first set of overlapping strips of material includes the front layer and the rear layer and does not include the support member.

12. The architectural covering of claim 11, wherein the intermediate layer of the second portion of the first overlapping strip of material of the first set of overlapping strips of material comprises a blackout portion so that the second portion includes the front layer, the rear layer, and the blackout

material.

13. The architectural covering of claim 11, wherein the support member comprises a thermoformable material.

14. The architectural covering of claim 11, wherein a front surface of the top end portion of the second elongated overlapping strip of material of the second set of overlapping strips of material is coupled to a rear surface of the bottom end portion of the first elongated overlapping strip of material of the second set of overlapping strips of material at a first coupling line.

15. The architectural covering of claim 14, wherein the first portion of the first elongated overlapping strip of material of the first set of overlapping strips of material is coupled to a front surface of the bottom end portion of the first elongated overlapping strip of material of the second set of overlapping strips of material proximate the first coupling line.

16. The architectural covering of claim 15, wherein, in use, the first portion of the first elongated overlapping strips of material of the first set of overlapping strips of material include a second fold positioned between the first coupling line and the first fold.
